1 Lecture 1 : Introduction to Machine Learning

This lecture was an introduction to machine learning. ...

2 Lecture 2

2.1 Solving Least Squares in General (for Linear models)

Suppose you are given a series of observations: $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ $X_{plot} = [0, 0.1, ..., 2]$ $Curve$

- Why noise ?
  - Since observations are not perfect
  - Owing to quantization / precision / rounding

Curve Fitting

Learn $f : X \to Y$ such that $E(f, X, Y_1)$ is minimized. Here the error function E and form of the function to learn f is chosen by the modeler.

Consider one such form of f,

$$f(x) = w_0 + w_1 x + w_2 x^2 + ... + w_t x^t$$

The sum of squares error is given by,

$$E = \frac{1}{2} \sum_{i=1}^{m} (f(x_i) - y_i)^2$$

So the expression is,

$$\underset{w=[w_1, w_2, ... w_t]}{\operatorname{argmin}} \frac{1}{2} \sum_{i=1}^{K} [(w_0 + w_1 x + w_2 x^2 + ... + w_t x^t) - y_1(i)]^2$$

If there are m data points, then a polynomial of degree $m-1$ can exactly fit the data, since the polynomial has m degrees of freedom (where degrees of freedom=no. of coefficients)

As the degree of the polynomial increases beyond m, the curve becomes more and more wobbly, while still passing through the points. Contrast the degree 10 fit in Figure 2.1 against the degree 5 fit in Figure 2.1. This is due to the problem of overfitting (overspecification)

Now E is a convex function. To optimize it, we need to set $\nabla_w E = 0$. The ∇ operator is also called gradient.

Solution is given by,

$$X = (\phi^t \phi)^{-1} \phi^t Y$$

If $m << t$ then

- ϕ becomes singular and the solution cannot be found OR
- The column vectors in ϕ become nearly linearly dependent

RMS (root mean sqare) error is given by :

$$RMS = \sqrt{\frac{2E}{k}}$$

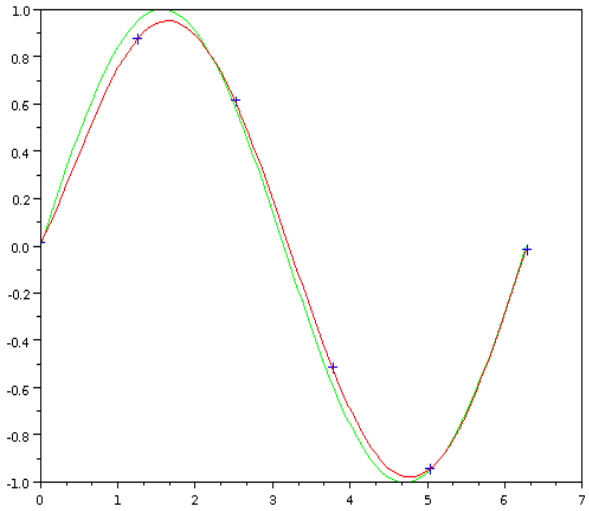


Figure 1: Fit for degree 5 polynomial.

Generally, some test data (which potentially could have been part of the training data) is held out for evaluating the generalized performance of the model. Another held out fraction of the training data, called the validation dataset is typically used to find the most appropriate degree t_{best} for f.

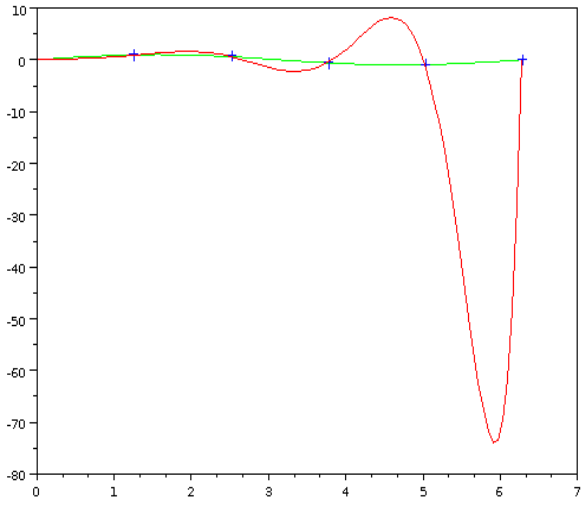


Figure 2: Fit for degree 10 polynomial. Note how wobbly this fit is.

3 Lecture 3 : Regression

This lecture was about regression. It started with formally defining a regression problem. Then a simple regression model called linear regression was discussed. Different methods for learning the parameters in the model were next discussed. It also covered least square solution for the problem and its geometrical interpretation.

3.1 Regression

Suppose there are two sets of variables $\mathbf{x} \in \Re^n$ and $\mathbf{y} \in \Re^k$ such that $\mathbf{x}$ is independent and y is dependant. The regression problem is concerned with determining y in terms of $\mathbf{x}$. Let us assume that we are given m data points $\mathcal{D} = \langle \mathbf{x}_1, \mathbf{y}_1 \rangle, \langle \mathbf{x}_2, \mathbf{y}_2 \rangle, .., \langle \mathbf{x}_m, \mathbf{y}_m \rangle$. Then the problem is to determine a function f^* such that $f^*(\mathbf{x})$ is the best predictor for $\mathbf{y}$, with respect to $\mathcal{D}$. Suppose $\varepsilon(f, \mathcal{D})$ is an error function, designed to reflect the discrepancy between the predicted value $f(\mathbf{x}')$ of $\mathbf{y}'$ and the actual value $\mathbf{y}'$ for any $\langle \mathbf{x}', \mathbf{y}' \rangle \in \mathcal{D}$, then

$$f^* = \underset{f \in \mathcal{F}}{\operatorname{argmin}}\ \varepsilon(f, \mathcal{D}) \tag{1}$$

where, $\mathcal{F}$ denotes the class of functions over which the optimization is performed.

3.2 Linear regression

Depending on the function class we consider, there are many types of regression problems. In Linear regression we consider only linear functions, functions that are linear in the basis function. Here $\mathcal{F}$ is of the form $\{\sum_{i=1}^{p} w_i \phi_i(\mathbf{x})\}$. $\phi_i : \mathbb{R}^n \to \mathbb{R}^k$ Here, the ϕ_i's are called the **basis functions** (for example, we can consider $\phi_i(x) = x^i$, *i.e.*, polynomial basis functions) .

Any function in $\mathcal{F}$ is characterized by its parameters, the w_i's. Thus, in (1) we have to find $f(\mathbf{w}^*)$ where

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}} \varepsilon(\mathbf{w}, \mathcal{D})$$

3.3 Least square solution

The error function ε plays a major role in the accuracy and tractability of the optimization problem. The error function is also called the **loss function**. The squared loss is a commonly used loss function. It is the sum of squares of the differences between the actual value and the predicted value.

$$\varepsilon(f, \mathcal{D}) = \sum_{\langle \mathbf{x}_i, y_i \rangle \in \mathcal{D}} (f(\mathbf{x}_i) - y_i)^2$$

So the least square solution for linear regression is given by

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}} \sum_{j=1}^{m} \Big(\sum_{i=1}^{p} (w_i \phi_i(x_j) - y_j \Big)^2$$

The minimum value of the squared loss is zero. Is it possible to achieve this value ? In other words is $\forall j,\ \sum_{i=1}^{p} w_i \phi_i(x_j) = y_j$ possible ?

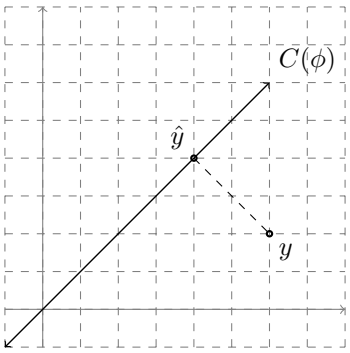


Figure 3: Least square solution $\hat{y}$ is the orthogonal projection of y onto column space of ϕ

The above equality can be written as $\forall u,\ \phi^T(x_u)\mathbf{w} = y_u$
or equivalently $\phi\mathbf{w} = \mathbf{y}$ where

$$\phi = \begin{bmatrix} \phi_1(\mathbf{x}_1) & \cdots & \phi_p(\mathbf{x}_1) \\ \vdots & \vdots & \vdots \\ \phi_1(\mathbf{x}_m) & \cdots & \phi_p(\mathbf{x}_m) \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

It has a solution if $\mathbf{y}$ is in the column space (the subspace of $\mathbb{R}^n$ formed by the column vectors) of ϕ. It is possible that there exists no $\mathbf{w}$ which satisfies the conditions? In such situations we can solve the least square problem.

3.4 Geometrical interpretation of least squares

Let $\hat{\mathbf{y}}$ be a solution in the column space of ϕ. The least squares solution is such that the distance between $\hat{\mathbf{y}}$ and $\mathbf{y}$ is minimized. From the diagram it is clear that for the distance to be minimized, the line joining $\hat{\mathbf{y}}$ to $\mathbf{y}$ should be orthogonal to the column space. This can be summarized as

1. $\phi\mathbf{w} = \hat{\mathbf{y}}$
2. $\forall v \in \{1, ..p\},\ (\mathbf{y} - \hat{\mathbf{y}})^T\phi_v = 0$ or $(\hat{\mathbf{y}} - \mathbf{y})^T\phi = 0$

$$\begin{aligned} \hat{\mathbf{y}}^T\phi &= \mathbf{y}^T\phi \\ ie,\ (\phi\mathbf{w})^T\phi &= \mathbf{y}^T\phi \\ ie,\ \mathbf{w}^T\phi^T\phi &= \mathbf{y}^T\phi \\ ie,\ \phi^T\phi\mathbf{w} &= \phi^T\mathbf{y} \\ \therefore\ \mathbf{w} &= (\phi^T\phi)^{-1}\mathbf{y} \end{aligned}$$

In the last step, please note that, $\phi^T\phi$ is invertible only if ϕ has full column rank.

Theorem: If ϕ has full column rank, $\phi^T\phi$ is invertible. A matrix is said to have full column rank if all its column vectors are linearly independent. A set of vectors $\mathbf{v}_i$ is said to be linearly independent if $\sum_i \alpha_i \mathbf{v}_i = 0 \Rightarrow \alpha_i = 0$.

Proof: Given that ϕ has full column rank and hence columns are linearly independent, we have that $\phi\mathbf{x} = 0 \Rightarrow \mathbf{x} = \mathbf{0}$.

Assume on the contrary that $\phi^T\phi$ is non invertible. Then $\exists \mathbf{x} \neq \mathbf{0} \ni \phi^T\phi\mathbf{x} = \mathbf{0}$.

$\Rightarrow \mathbf{x}^T\phi^T\phi\mathbf{x} = 0$

$\Rightarrow (\phi\mathbf{x})^T\phi\mathbf{x} = ||\phi\mathbf{x}||^2 = 0$

$\Rightarrow \phi\mathbf{x} = \mathbf{0}$. This is a contradiction. Hence the theorem is proved.

4 Lecture 4 : Least Squares Linear Regression

In this lecture we discussed how to minimize the error function $\varepsilon(\mathbf{w}, \mathcal{D})$ that we used for the least square linear regression model in the last lecture. To do this, some basic concepts regarding minimization of a function were discussed and we applied these to our error function.

4.1 Least Square Linear Regression Model

In the least squares regression model, we determine the value of $\mathbf{w}$ for which our error function ε attains the minimum value. Here, $\mathcal{D} =< \mathbf{x}_1, y_1 >, < \mathbf{x}_2, y_2 >, .., < \mathbf{x}_m, y_m >$ is the training data set, and ϕ_i's are the basis functions.

$$\begin{aligned}
\mathbf{w}^* &= \underset{\mathbf{w}}{\operatorname{argmin}} \left\{ \sum_{j=1}^{m} \left(f\left(\mathbf{x}_j, \mathbf{w}\right) - y_j \right)^2 \right\} \\
&= \underset{\mathbf{w}}{\operatorname{argmin}} \left\{ \sum_{j=1}^{m} \left(\sum_{i=1}^{p} w_i \phi_i(\mathbf{x}_j) - y_j \right)^2 \right\}
\end{aligned}$$

$$\phi = \begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & ..... & \phi_p(\mathbf{x}_1) \\ & . & & \\ & . & & \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & ..... & \phi_p(\mathbf{x}_m) \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ . \\ . \\ y_m \end{bmatrix}$$

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ . \\ . \\ w_p \end{bmatrix}$$

$$\begin{aligned}
\varepsilon &= \min_{\mathbf{w}} \sum_{j=1}^{m} \left(\phi^T(\mathbf{x}_j)\mathbf{w} - y_j \right)^2 \\
&= \min_{\mathbf{w}} ||\phi\mathbf{w} - \mathbf{y}||^2 \\
&= \min_{\mathbf{w}} \left(\phi\mathbf{w} - \mathbf{y}\right)^T (\phi\mathbf{w} - \mathbf{y}) \\
&= \min_{\mathbf{w}} \left(\mathbf{w}^T \phi^T \phi \mathbf{w} - 2\mathbf{y}^T \phi \mathbf{w} + \mathbf{y}^T \mathbf{y} \right) \qquad (2)
\end{aligned}$$

How to minimize a function?

Following are some basic concepts which help in minimizing or maximizing a function:

4.2 Level Curves and Surfaces

A *level curve* of a function $f(\mathbf{x})$ is defined as a curve along which the value of the function remains unchanged while we change the value of it's argument $\mathbf{x}$. Note that there can be as many level curves for any function as the number of different values it can attain.

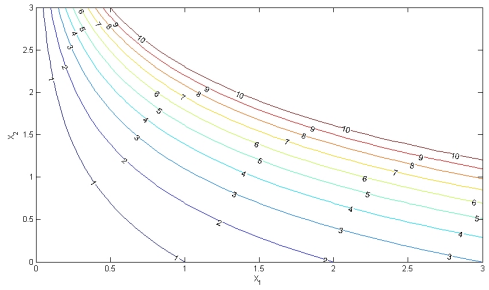


Figure 4: 10 level curves for the function $f(x_1, x_2) = x_1 e_2^x$ (Figure 4.12 from [1])

Level surfaces are similarly defined for any n-dimensional function $f(x_1, x_2, ..., x_n)$ as a collection of points in the argument space on which the value of the function is same at all points while we change the argument values.

Formally we can define a level curve as :

$$L_c(f) = \left\{ \mathbf{x} | f(\mathbf{x}) = c \right\}$$

where c is a constant.

Refer to Fig. 4.15 in class notes [1] for example.

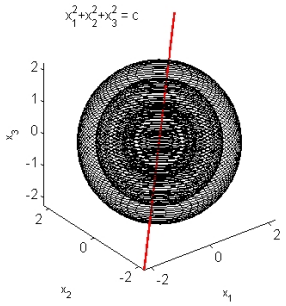


Figure 5: 3 level surfaces for the function $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$ with $c = 1, 3, 5$. The gradient at $(1, 1, 1)$ is orthogonal to the level surface $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 = 3$ at $(1, 1, 1)$ (Fig. 4.14 from [1]).

4.3 Gradient Vector

The *gradient vector* of a function f at a point $\mathbf{x}$ is defined as follows:

$$\nabla f_{\mathbf{x}^*} = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ . \\ . \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} \epsilon \mathbb{R}^n$$

The direction of the gradient vector gives the direction of maximum rate of change of the value of the function at a point. Also the magnitude of the gradient vector gives that maximum value of rate of change.

Refer to Definition 23 in the class notes [1] for more details.

4.4 Directional Derivative

Directional Derivative gives the rate of change of the function value in a given direction at a point. The *directional derivative* of a function f in the direction of a unit vector $\mathbf{v}$ at a point $\mathbf{x}$ can be defined as :

$$D_{\mathbf{v}}(f) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h}$$

$$||\mathbf{v}|| = 1$$

Note: The maximum value of directional derivative of a function f at any point is always the magnitude of it's gradient vector at that point.

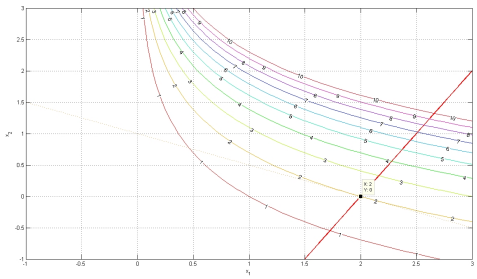


Figure 6: The level curves from Figure 4 along with the gradient vector at (2, 0). Note that the gradient vector is perpenducular to the level curve $x_1 e^{x_2} = 2$ at (2, 0) (Figure 4.13 from [1])

Refer to Definition 22 and Theorem 58 in the class notes [1] for more details.

4.5 Hyperplane

A Hyperplane is a set of points whose direction *w.r.t.* a point $\mathbf{p}$ is orthogonal to a vector $\mathbf{v}$. It can be formally defined as :

$$H_{\mathbf{v},\mathbf{p}} = \left\{ \mathbf{q} \mid (\mathbf{p} - \mathbf{q})^T \mathbf{v} = 0 \right\}$$

4.6 Tangential Hyperplane

There are two definitions of *tangential hyperplane* ($TH_{\mathbf{x}^*}$) to *level surface* ($L_{f(\mathbf{x}^*)}(f)$) of f at $\mathbf{x}^*$:

1. Plane consisting of all tangent lines at $\mathbf{x}^*$ to any parametric curve $c(t)$ on level surface.

2. Plane orthogonal to the gradient vector at $\mathbf{x}^*$.

$$TH_{\mathbf{x}^*} = \left\{ \mathbf{p} \mid (\mathbf{p} - \mathbf{x}^*)^T \nabla f(\mathbf{x}^*) = 0 \right\}$$

Note: By definition, $TH_{\mathbf{x}^*}$ is $n-1$ dimensional.

Refer to Definition 24 and Theorem 59 in class notes [1] for more details.

4.7 Gradient Descent Algorithm

Gradient Descent Algorithm is used to find minimum value attained by a real valued function $f(\mathbf{x})$. We first start at an intial point $\mathbf{x}^{(0)}$ and make a sequence of steps proportional to negative of gradient of the function at the point. Finally we stop at a point $\mathbf{x}^{(*)}$ where a desired convergence

criterion (see notes on Convex Optimization) will be attained.

The idea of gradient descent algorithm is based on the fact that if a real-valued function $f(\mathbf{x})$ is defined and differentiable at a point $\mathbf{x}^k$, then $f(\mathbf{x})$ decreases fastest when you move in the direction of the negative gradient of the function at that point, which is $-\nabla f(\mathbf{x})$.

Here we describe the method of Gradient Descent Algorithm to find the *parameter vector* $\mathbf{w}$ which minimizes the error function, $\varepsilon(\mathbf{w}, \mathcal{D})$

$$\begin{aligned}
\Delta\mathbf{w}^{(k)} &= -\nabla\varepsilon(w^{(k)}) \qquad \textbf{from equation } (2) \\
&= -\nabla(\mathbf{w}^T\phi^T\phi\mathbf{w} - 2\mathbf{y}^T\phi\mathbf{w} + \mathbf{y}^T\mathbf{y}) \\
&= -(2\phi^T\phi\mathbf{w} - 2\mathbf{y}^T\phi + 0) \\
&= 2(\phi^T\mathbf{y} - \phi^T\phi\mathbf{w})
\end{aligned}$$

so we got

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} + 2t^{(k)}(\phi^T\mathbf{y} - \phi^T\phi\mathbf{w}^{(k)})$$

Gradient Descent Algorithm :

Find starting point $\mathbf{w}^{(0)}\epsilon\mathcal{D}$
repeat
1. $\Delta\mathbf{w}^k = -\nabla\varepsilon(\mathbf{w}^{(k)})$
2. Choose a step size $t^{(k)} > 0$ using exact or backtracking ray search.
3. Obtain $\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} + t^{(k)}\Delta\mathbf{w}^{(k)}$.
4. Set $k = k + 1$.

until stopping criterion (such as $\| \nabla\varepsilon(\mathbf{x}^{(k+1)}) \| \leq \epsilon$) is satisfied

Exact Line Search Algorithm :

$$t^{(k)} = \underset{t}{\operatorname{argmin}}\varepsilon\left(\mathbf{w}^{(k)} + 2t\left(\phi^T\mathbf{y} - \phi^T\phi\mathbf{w}^{(k)}\right)\right)$$

In general

$$t^{(k)} = \underset{t}{\operatorname{argmin}} f\left(\mathbf{w}^{(k+1)}\right)$$

Refer to section 4.5.1 in the class notes [1] for more details.

4.8 Local Minimum and Local Maximum

Critical Point : $\mathbf{x}$ is a called a *critical point w.r.t* to a function f, if $\nabla f(\mathbf{x}) = 0$ i.e. the gradient vanishes at $\mathbf{x}$ or the gradient fails to exist at $\mathbf{x}$.

Local Minimum (or Maximum):

If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ then $\mathbf{x}^*$ can be a point of local minimum (or maximum). [*Neccessary Condition*]

If $\nabla^2 f(\mathbf{x}^*)$ is positive (negative) definite then $\mathbf{x}^*$ is a point of local minimum (maximum). [*Sufficient Condition*]

Note: $\nabla^2 f(\mathbf{x}^*)$ is positive definite means :

$$\forall \mathbf{x} \neq \mathbf{0} \quad \mathbf{x}^T \nabla^2 f(\mathbf{x}^*) \mathbf{x} > \mathbf{0}$$

OR

$$\lambda_i(\nabla^2 f(\mathbf{x}^*)) > 0$$

i.e. matrix eigen values are positive.

Refer to definition 27, theorem 61 and fig. 4.23, 4.24 in the class notes [1] for more details.

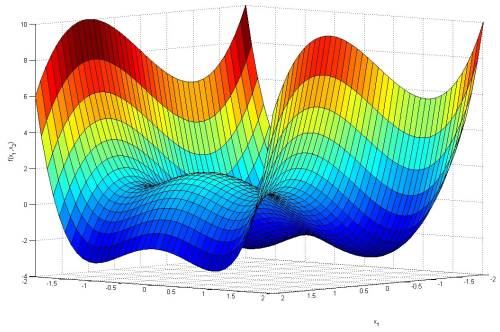


Figure 7: Plot of $f(x_1, x_2) = 3x_1^2 - x_1^3 - 2x_2^2 + x_2^4$, showing the various local maxima and minima of the function (fig. 4.16 from [1])

5 Lecture 5 : Convex functions

In this lecture the concepts of convex sets and functions were introduced.

5.1 Recap

We recall that the problem was to find $\mathbf{w}$ such that

$$\begin{aligned} \mathbf{w}^* &= \underset{\mathbf{w}}{\text{argmin}} \, ||\phi\mathbf{w} - \mathbf{y}||^2 && (3) \\ &= \text{argmin}_{\mathbf{w}}(\mathbf{w}^T\phi^T\phi\mathbf{w} - 2\mathbf{w}^T\phi\mathbf{y} - \mathbf{y}^T\mathbf{y}) && (4) \end{aligned}$$

5.2 Point 1

If $\nabla f(\mathbf{x}^*)$ is defined & $\mathbf{x}^*$ is local minimum/maximum, then $\nabla f(\mathbf{x}^*) = 0$
(A necessary condition) `(Cite : Theorem 60)[2]`

Given that

$$\begin{aligned} f(\mathbf{w}) &= \underset{\mathbf{w}}{\text{argmin}}(\mathbf{w}^T\phi^T\phi\mathbf{w} - 2\mathbf{w}^T\phi\mathbf{y} - \mathbf{y}^T\mathbf{y}) && (5) \\ \implies \nabla f(\mathbf{w}) &= 2\phi^T\phi\mathbf{w} - 2\phi^T\mathbf{y} && (6) \end{aligned}$$

we would have

$$\begin{aligned} \nabla f(\mathbf{w}^*) &= 0 && (7) \\ \implies 2\phi^T\phi\mathbf{w}^* - 2\phi^T\mathbf{y} &= 0 && (8) \\ \implies \mathbf{w}^* &= (\phi^T\phi)^{-1}\phi^T\mathbf{y} && (9) \end{aligned}$$

5.3 Point 2

Is $\nabla^2 f(\mathbf{w}^*)$ positive definite ?
i.e. $\forall \mathbf{x} \neq 0$, is $\mathbf{x}^T\nabla f(\mathbf{w}^*)\mathbf{x} > 0$? (A sufficient condition for local minimum)

(Note : Any positive definite matrix is also positive semi-definite)
`(Cite : Section 3.12 & 3.12.1)[3]`

$$\begin{aligned} \nabla^2 f(\mathbf{w}^*) &= 2\phi^T\phi && (10) \\ \implies \mathbf{x}^T\nabla^2 f(\mathbf{w}^*)\mathbf{x} &= 2\mathbf{x}^T\phi^T\phi\mathbf{x} && (11) \\ &= 2(\phi\mathbf{x})^T\phi\mathbf{x} && (12) \\ &= 2\,||\phi\mathbf{x}||^2 \geq 0 && (13) \end{aligned}$$

And if ϕ ***has full column rank*** ,

$$\phi\mathbf{x} = 0 \quad iff \quad \mathbf{x} = 0 \tag{14}$$

$\therefore$ If $\mathbf{x} \neq 0, \quad \mathbf{x}^T \nabla^2 f(\mathbf{w}^*)\mathbf{x} > 0$
Example where ϕ doesn't have a full column rank,

$$\phi = \begin{bmatrix} x_1 & x_1^2 & x_1^2 & x_1^3 \\ x_2 & x_2^2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ x_n & x_n^2 & x_n^2 & x_n^3 \end{bmatrix} \tag{15}$$

This is the simplest form of linear correlation of features, and it is not at all desirable.

5.4 Point 3

Definition of convex sets and convex functions `(Cite : Definition 32 and 35)[2]`

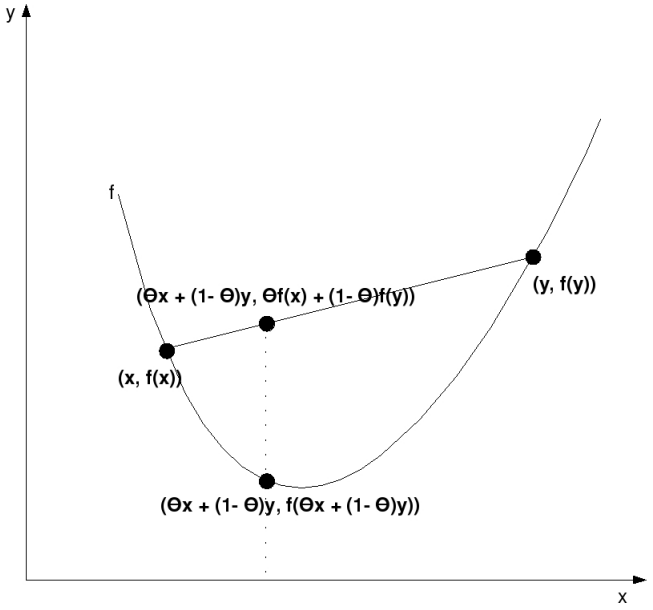


Figure 8: A sample convex function

$$\therefore f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}) \tag{16}$$

Some convex functions : `(Cite : Table 4.1, pg-54)[2]`

To prove : *Verify that a hyperplane is a convex set.*
Proof :

A Hyperplane $\mathcal{H}$ is defined as $\{\mathbf{x}|\mathbf{a}^T\mathbf{x} = b, \mathbf{a} \neq \mathbf{0}\}$. Let $\mathbf{x}$ and $\mathbf{y}$ be vectors that belong to the hyperplane. Since they belong to the hyperplane, $\mathbf{a}^T\mathbf{x} = b$ *and* $\mathbf{a}^T\mathbf{y} = b$. In order to prove the convexity of the set we must show that :

$$\theta\mathbf{x} + (1-\theta)\mathbf{y} \in \mathcal{H},\ where\ \theta \in [0,1] \tag{17}$$

In particular, it will belong to the hyperplane if it's true that :

$$\begin{aligned} \mathbf{a}^T(\theta\mathbf{x} + (1-\theta)\mathbf{y}) &= b && (18)\\ \implies \mathbf{a}^T\theta\mathbf{x} + \mathbf{a}^T(1-\theta)\mathbf{y} &= b && (19)\\ \implies \theta\mathbf{a}^T\mathbf{x} + (1-\theta)\mathbf{a}^T\mathbf{y} &= b && (20)\\ & && (21) \end{aligned}$$

And, we also have $\mathbf{a}^T\mathbf{x} = b$ *and* $\mathbf{a}^T\mathbf{y} = b$. Hence $\theta b + (1-\theta)b = b$. [Hence Proved]

So a hyperplane is a convex set.

Q. *Is* $||\phi\mathbf{w} - \mathbf{y}||^2$ *convex?*
A. To check this, we have `(Cite : Theorem 75)[2]` but it is not very practical. We would use `(Cite : Theorem 79)[2]` to check for the convexity of our function. So the condition that has our focus is -

$$\nabla^2 f(\mathbf{w}^*)\ is\ positive\ semi-definite,\ if\ \forall\mathbf{x} \neq 0,\ \mathbf{x}^T\nabla^2 f(\mathbf{w}^*)\mathbf{x} \geq 0 \tag{22}$$

We have,

$$\nabla^2 f(\mathbf{w}) = 2\phi^T\phi \tag{23}$$

So, $||\phi\mathbf{w} - \mathbf{y}||^2$ is convex, since the domain for $\mathbf{w}$ is $\mathbb{R}^n$ and is convex.

Q. *Is* $||\phi\mathbf{w} - \mathbf{y}||^2$ *strictly convex?*
A. Iff ϕ has full column rank.

(Side note : Weka[1])

5.5 Point 4

To prove: *If a function is convex, any point of local minima* $\equiv$ *point of global minima*
Proof - `(Cite : Theorem 69)[2]`

To prove : *If a function is strictly convex, it has a unique point of global minima*
Proof - `(Cite : Theorem 70)[2]`

Since $||\phi\mathbf{w} - \mathbf{y}||^2$ is strictly convex for linearly independent ϕ,

$$\nabla f(\mathbf{w}^*) = 0\ for\ \mathbf{w}^* = (\phi^T\phi)^{-1}\phi^T\mathbf{y} \tag{24}$$

[1] `http://www.cs.waikato.ac.nz/ml/weka/`